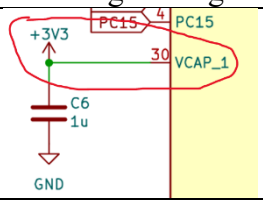
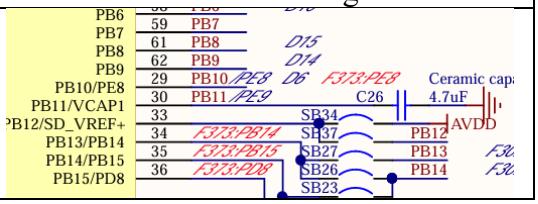
Agricultural VCU V2 Version Control Report

1. Overview

Version	Date	Status
V1	2025-02	Functioning with conditions[1]
V2	2026-01	On Going (PCBA)[2]

[1] The PCB V1 has a schematic mistake on VCAP routing; one of the pins should not be connected to 3V3. As this was noticed after manufacturing, the trace had to be cut manually.

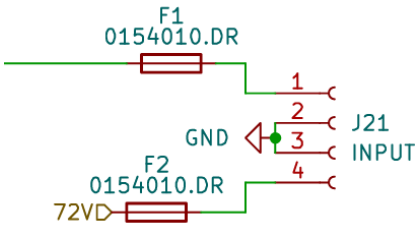
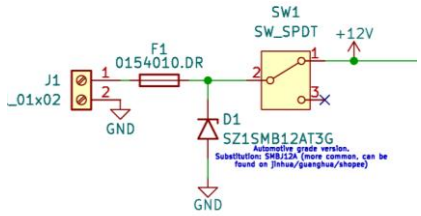
Wrong routing	Correct routing
	

[2] The pinout order for the relay terminal block was flipped

Current PCB pinout order (from left)	Correct pinout order (from left)
NC – COM - NO	NO – COM - NC

2. Schematic

2.1. Power Supply

V1	V2
Terminal block series number: NC6-P107-04	Terminal block series number: NC6-P107-02
	

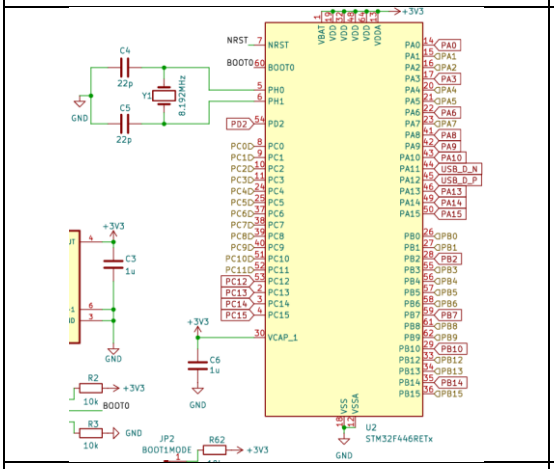
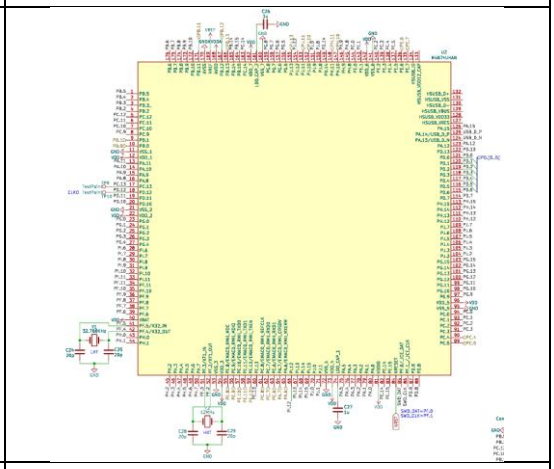
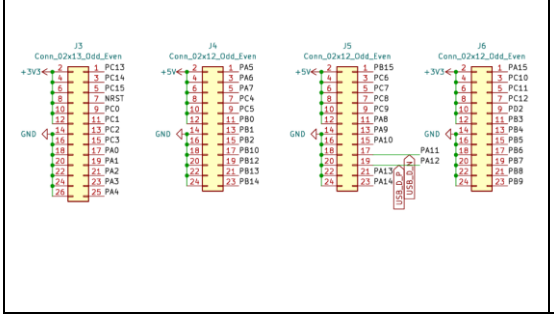
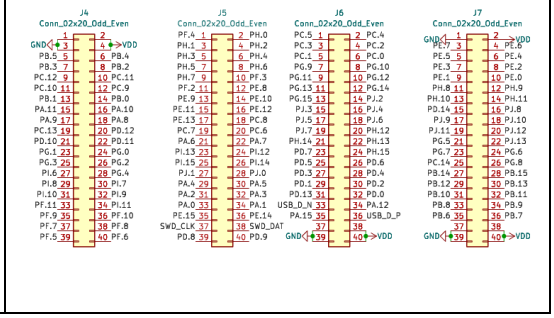
The power supply part in V2 adapts from V1, featuring a change in the terminal block used for input voltage and additional protection with a unipolar TVS (transient voltage suppressor). In the previous version, a 72V supply was needed for two of the relays, so the terminal block used has 4 pins: 12V rail, 72V rail, and ground for each voltage rail. In the current version, a 72V supply is no longer needed. To reduce space used, the terminal block for input voltage is changed to the 2-pin version of the same series. In PCB V2, the power supply has additional protection using a unipolar TVS diode. TVS diode protects the voltage rail from sudden voltage surges. There are two options for the TVS diode which have the same specifications, but different grading of the component. SZ1SMB12AT3G is preferred if a higher level of reliability is required, because it is an automotive-grade TVS diode. If availability is more prioritised, it can be changed to SMB12A, a general-purpose version of SZ1SMB12AT3G. The comparison is in the table below:

Feature	SZ1SMB12AT3G	SMB12A
Manufacturer	Littelfuse (or Onsemi)	Littelfuse (or others like STMicro, etc.)
Qualification	AEC-Q101 (Automotive)	General purpose

Peak Pulse Power	600 W	600 W
Reverse Standoff Voltage	12 V	12 V
Breakdown Voltage (Min)	13.3 V	13.3 V
Clamping Voltage (Max)	19.9 V	19.9 V
Polarity	Unidirectional	Unidirectional
Package	DO-214AA (SMB)	DO-214AA (SMB)

2.2. Microcontroller Unit (MCU) Circuit

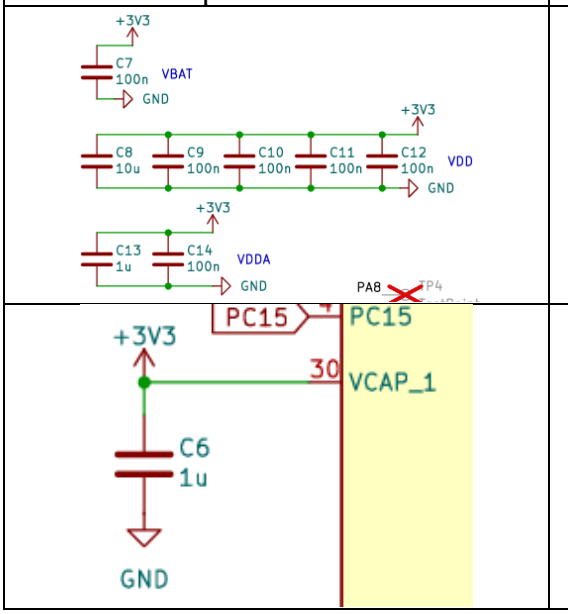
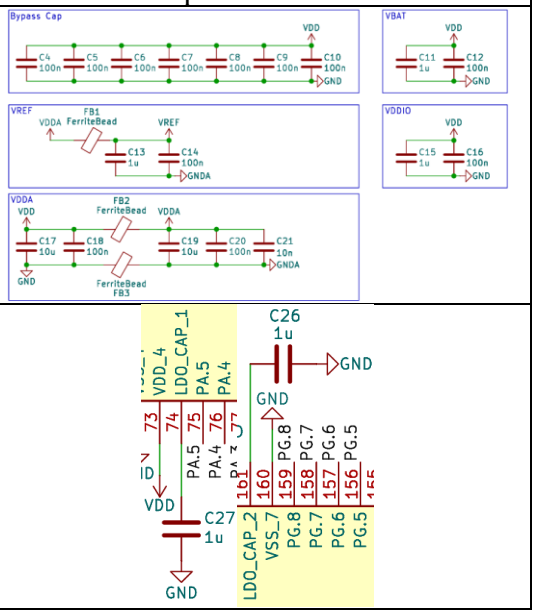
2.2.1. Microcontroller IC and GPIO Extension Pins

V1	V2
STM32F446RE	Nuvoton M467HJHAN
	
	

In V1, the control unit is based on STM32F446RE, an Arm® 32-bit Cortex®-M4 MCU with a footprint of LQFP64. The external clock used is an 8.192 MHz crystal with load capacitor pairs of 22pF. PCB V2 uses an MCU with the same core, but different manufacturers and footprints. The MCU used is Nuvoton M467HJHAN, whose footprint is LQFP176, which provides more pinouts for GPIO. Referring to the Numaker M467 schematic, PCB V2 MCU circuit uses a 12MHz crystal for external clock, with load capacitor pairs of 20pF. An external RTC clock is optional for both MCUs. PCB V1 does not use that feature, but PCB V2 uses an external RTC clock. An external clock is used because it is more accurate and stable across temperature and voltage changes.

Each extension pin in PCB V1 has one side for GPIO and another side for power, 2 connectors provide 3V3 and 2 connectors provide 5V. The circuit uses one 2x13 pin socket and 3 2x12 pin sockets. In PCB V2, there are many more GPIO pins, but the area in the PCB layout is limited. So, PCB V2 uses 4 pieces 2x20 pin socket for extension pins connector with reduced pins for power and the voltage provided from the connectors are limited to 3V3.

2.2.2. Decoupling Capacitors

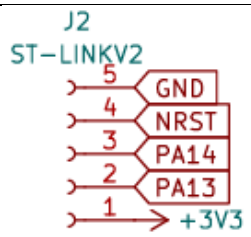
V1	V2
Footprint used: 0402	Footprint used: 0603
	

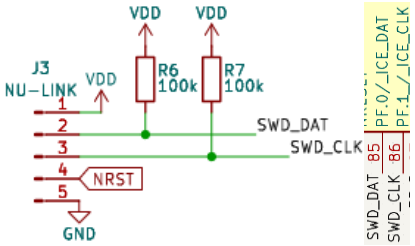
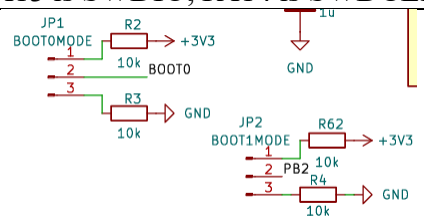
In the previous version (V1), the footprint used for all decoupling capacitors was 0402. It was chosen to give more space for routing to the GPIO extension pins. However, those capacitors are difficult to solder because the footprint is tiny. Additionally, the VCAP capacitor, which also uses a 0402 footprint, has a routing issue. The first pad of the capacitor should be connected only to the VCAP pin of the MCU; however, it is also connected to 3V3, which failed the 3V3 power rail. Fixing this issue requires precision in cutting the trace that connects the VCAP capacitor and 3V3. This is quite challenging due to the small trace and footprint. Therefore, in the current version (V2), the two VCAP pins are reassured that there is no connection between LDO_CAP pins and 3V3 power rail to prevent failure like in PCB V1. In addition, all decoupling capacitors' footprints are changed to 0603 in the current version, referring to Numaker M467 hardware reference and to ease the soldering process. The value of each decoupling capacitor in PCB V2 follows the schematic reference of Numaker M467.

Separating analogue and digital voltage supply in a microcontroller is optional. However, if it is applied in the reference circuit, it is recommended to separate the routing for analogue and digital voltage and their grounding. In the reference design, the analogue supply (AVDD) is isolated from the digital supply (VDD) using a ferrite bead to prevent switching noise between digital and analogue circuitry from coupling. In addition, the analogue ground (VSSA) is connected to the digital ground through a ferrite bead, creating a locally quiet analogue

reference region while still maintaining a defined DC ground reference through the continuous ground plane. This configuration attenuates high-frequency digital return currents from flowing through the analogue reference network while allowing low-frequency and DC return currents to flow normally. To avoid ground potential differences and common-mode errors in the ADC and DAC, the analogue components, VREF network, and decoupling capacitors are placed close to the MCU so that analogue return currents remain localised, and a solid ground plane is maintained under both domains to ensure a stable global reference.

2.2.3. Serial Wire Debug (SWD) and Booting Circuit

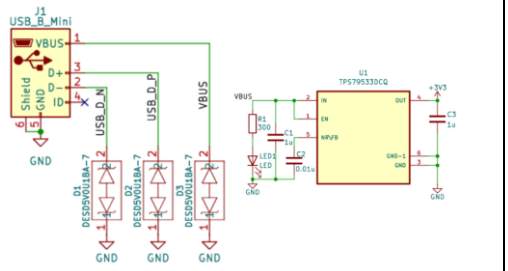
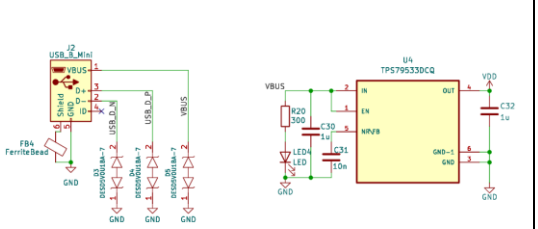
<table>
<tr><th>V1</th><th>V2</th></tr>
<tr><td> 

PA13 is SWDIO, PA14 is SWDCLK</td><td> 
</td></tr>
<tr><td> 
</td><td>Not used</td></tr>
</table>

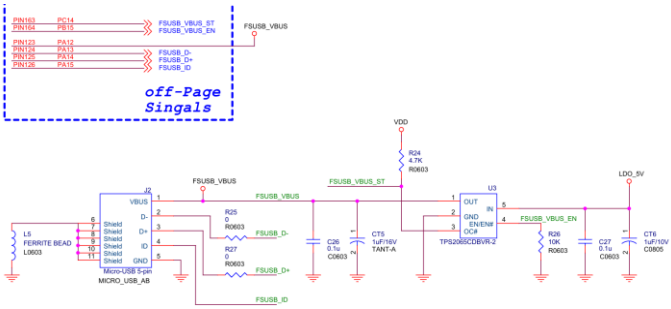
Serial wire debug (SWD) is an electrical alternative JTAG interface for programming ARM-based microcontrollers. While JTAG has various versions of pinouts used, SWD only requires 5-pin connections between host (debugger) and target (microcontroller). The necessary logic is carried by SWDIO pin, which is a bidirectional data line used for communication, and synchronised by clock signal driven by the host through SWCLK pin. There is also a nRESET or NRST pin to allow the debugger to reset the target during debug session. The other two pins are for 3V3 and ground. The GPIO pins used for SWDIO and SWDCLK are different for every microcontroller. In the STM32F446 used in V1, SWDIO and SWDCLK are mapped to PA13 and PA14, respectively. In the Nuvoton M467 used in V2, the corresponding pins are PF0 (SWD_DAT) and PF1 (SWD_CLK). The M467 requires external pull-up resistors on both SWD lines to guarantee stable logic-high levels during reset and debug entry. 100 kΩ pull-up resistors to VDD are added in the V2 schematic. The STM32F446 provides sufficient internal pull-ups on the SWD pins, so no external resistors were required in V1. The M467 design does not need the same explicit BOOT0/BOOT1 booting-logic jumper network seen in the STM32 version, simplifying the “boot mode” hardware around the MCU.

2.2.4. USB Circuit

V1	V2

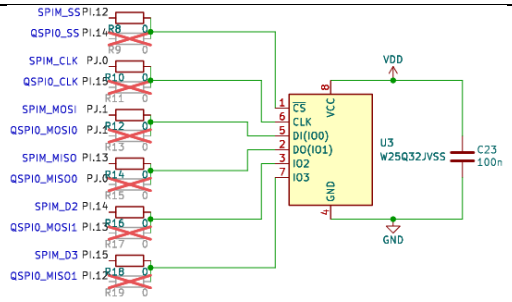
The schematic of USB circuit in V1 and V2 are the same but differ in how the shield pin of USB port is treated. In V1, the shield pin is connected directly to the ground, while in V2, the shield pin is isolated from the ground with ferrite bead. The ferrite bead creates a high-impedance path for high-frequency noise while still providing a DC reference, isolating the shield from digital ground for EMI and ESD currents.



Full-speed (FS) USB Schematic from Nuvoton reference

PCB V2 does not exactly follow the Nuvoton USB reference schematic because the board is designed as a USB device-only system rather than a full USB OTG or host-capable platform. The Nuvoton reference includes VBUS sensing and power-switch control (VBUS_ST and VBUS_EN) to support role switching between host and device, which requires monitoring and actively driving the 5 V VBUS line; however, in this design the MCU always operates as a USB device and never sources VBUS, so those functions are electrically unnecessary. Instead, V2 directly regulates the USB VBUS through a fixed LDO and focuses on signal integrity and EMC robustness by implementing proper ESD protection.

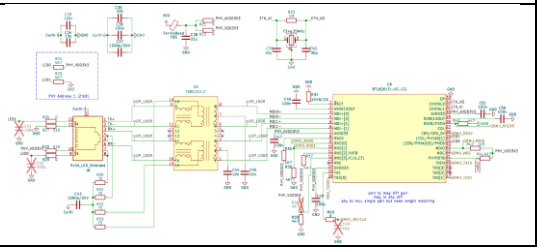
2.2.5. Flash Memory

V1	V2
Not used	

The inclusion of external flash memory in V2 represents an enhancement over V1's architecture, which did not employ non-volatile storage beyond the microcontroller's internal flash. V2 incorporates a serial SPI flash device

connected to dedicated SPI pins, providing additional code or data storage capacity that can be used for logging, bootloader storage, configuration memory, or runtime data buffering. SPI flash memory offers a low-pin-count, high-density, and cost-effective solution for expanding non-volatile memory, which is particularly advantageous in embedded systems requiring persistent data retention across power cycles and firmware upgrades. The selected device is configured with controlled trace routing and proper decoupling to maintain signal integrity at the target SPI clock rates, and pull-up resistors are used where recommended to ensure defined logic levels during reset or idle states. By integrating external flash, the design gains flexibility in managing firmware images and user data while maintaining compliance with recommended SPI timing and layout practices for robust communication between the microcontroller and memory device.

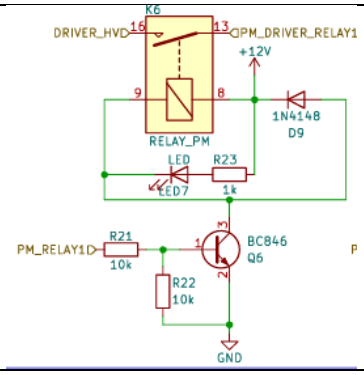
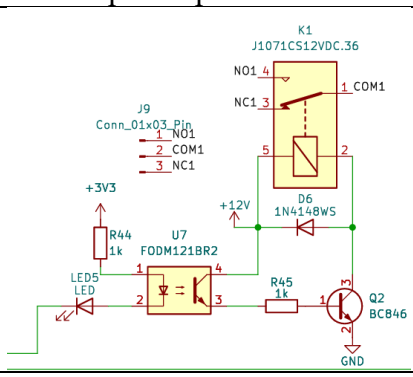
2.3. Ethernet

V1	V2
Not used	

In the previous version, the control unit communicated with mini-PC through UART, so Ethernet was not used. There are also no GPIO pins of STM32F446 that are dedicated for RMII. Meanwhile, M467 series MCU has GPIO pins dedicated to RMII. Moreover, communication through Ethernet is vastly faster than UART. Ethernet can send data in megabits and can be up to 10 Gbps, while UART can only send data in kilobits per second (Kbps) range. Therefore, an Ethernet circuit is applied in PCB V2 because Ethernet can provide faster communication between robot control unit and mini-PC.

2.4. Relays and GPIOs

2.4.1. Relay

V1	V2
Amounts:12	Amounts: 6
Connected to D-Sub ports for every 5 relays and a terminal block for each external relay	All relays are connected to terminal block for each relay
BJT driven	Optocoupler driven
	

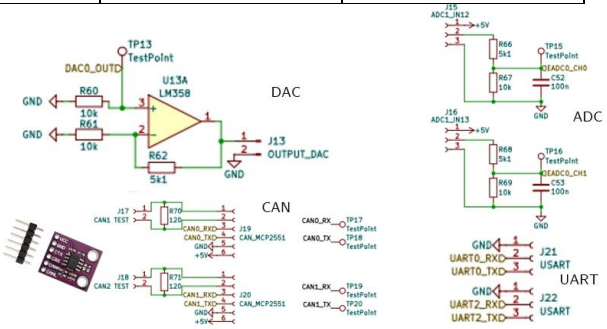
The previous version (V1) uses a BJT-based driven SPDT-NO 12 V relay, where the relay coil is switched by a transistor stage tied to the control logic. In this approach, the control ground and the relay coil ground are electrically connected, which is simple but exposes the MCU side to switching noise, ground bounce, and inductive transients generated when the relay is energised or released. Flyback protection is present, but fast current edges and EMI can still couple back into the logic domain, especially in an environment with long external cables or inductive loads.

In the current version (V2), the design is upgraded to an optocoupler-based SPDT 12 V relay circuit, introducing galvanic isolation between the control logic and the relay coil. The optocoupler cleanly separates the low-voltage MCU domain from the high-current 12 V relay domain, significantly improving noise immunity and protecting the MCU from voltage spikes and ground potential differences. The relay coil is driven on the isolated side with proper flyback suppression, while the control signal only drives the optocoupler LED, reducing stress on the MCU pins.

In terms of versatility, the output interface is also improved: the previous version relied on a D-sub connector, which is compact but fixed in pinout and less convenient for field wiring. The current version replaces this with a terminal block, allowing flexible cable types, easier installation, and simpler reconfiguration for different load wiring scenarios, making the board more suitable for industrial and field-deployment use cases.

2.4.2. GPIO

Interface	PCB V1	PCB V2
ADC	✓	✓
DAC	✓	✓
CAN	✓	✓
UART	✓ (1 port)	✓ (2 ports)
SPI (used for SD card)	✓	×
PWM	×	✓



Overall, the interfaces used remain the same in the previous and current versions. However, in the current version, the PWM interface is added for the motor driver and Ethernet for communication with the mini-PC through switch. Ports for UART interface is added one more from the previous version to give more options for UART in case it is needed. The current version has two PWM ports, one uses BPWM0_CH0 and the other uses BPWM1_CH0, using the same channel as the

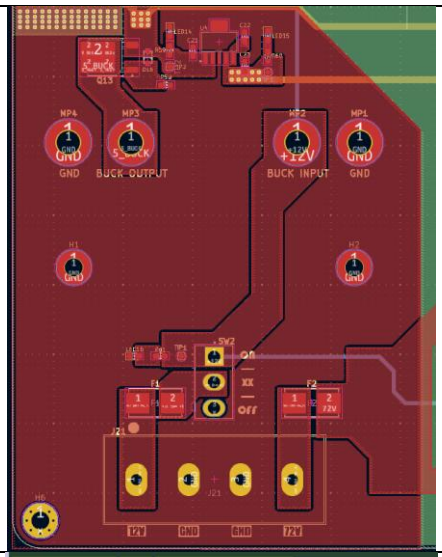
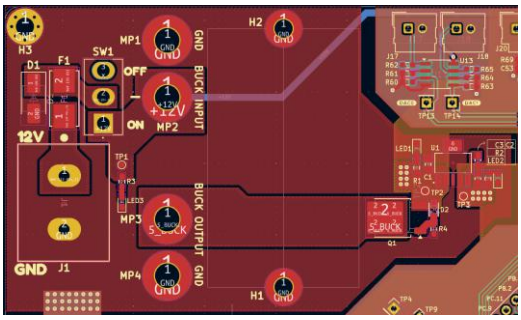
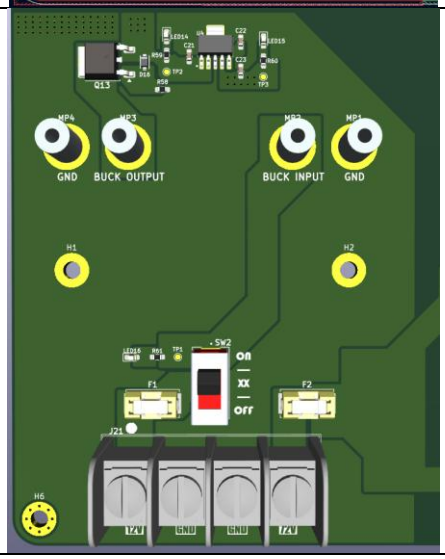
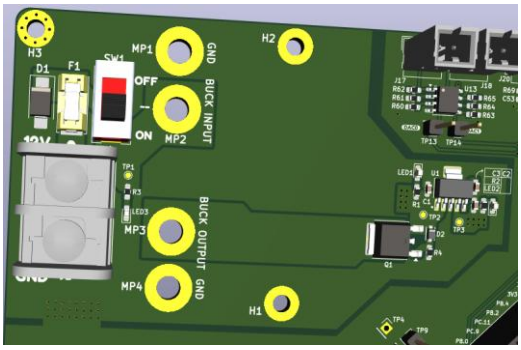
control unit used in UGV. The PWM circuit uses a transistor-based logic level shifter with BSS138, referring to the logic level module used in the UGV. SD card module is no longer used in the current version, so the circuit for SPI is omitted.

3. Layout

3.1. PCB Specifications

Specifications	PCB V1	PCB V2
Layer Count	4	4
Dimensions	180 mm x 180 mm	176.53 mm x 184.15 mm
PCB Thickness	2 mm	2 mm
Minimum track width	0.15 mm	0.1 mm
Minimum track spacing	0.15 mm	0.1 mm
Minimum hole diameter	0.3 mm	0.2 mm

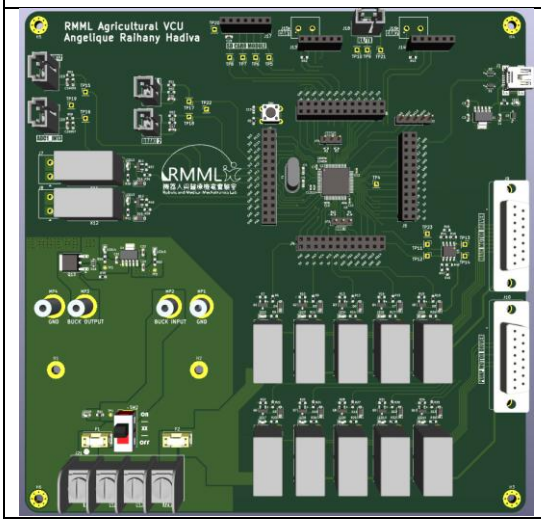
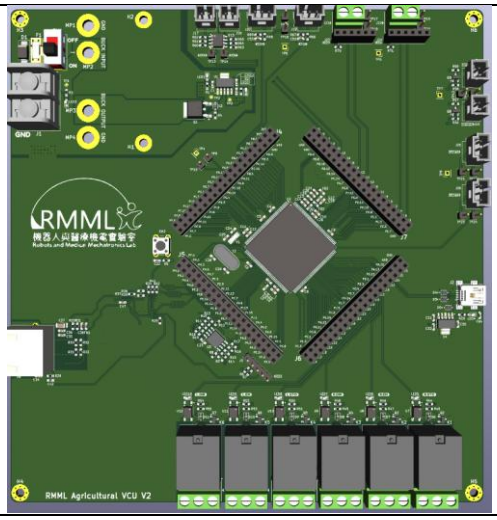
3.2. Power Supply

V1	V2
	
	

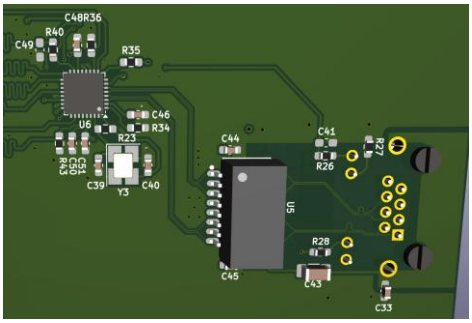
The power supply part of the layout is made more compact in the current version to give more space for GPIO and MCU circuits. Other changes made are terminal block selection and additional protection with a TVS diode. Previously, the fuses and switch were out in front of the terminal block. In the current version, as the voltage rail needed

is only 12V, the fuse needed is reduced to one piece. So, it is possible to make the input voltage area more compact.

3.3. Component placements

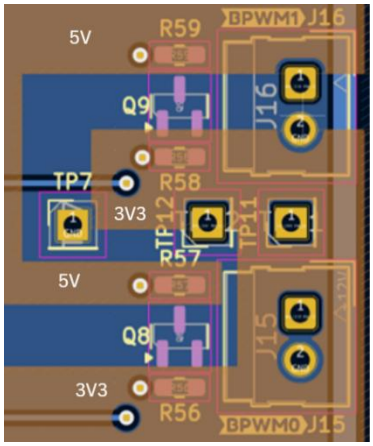
V1	V2
	

In PCB V1, all components are placed on the top layer to ease the soldering process. In PCB V2, it is possible to do the same, but would make the routing, specifically for ethernet part, more difficult. Referring to Numaker M467 layout, some components are placed on the bottom layer to give more space for length matching from ethernet PHY IC to the MCU.



Components of ethernet circuit which is placed on bottom layer

Interface	12V	5V	3V3
ADC		✓	
DAC	✓		
CAN		✓	
UART	-	-	-
PWM		✓	✓

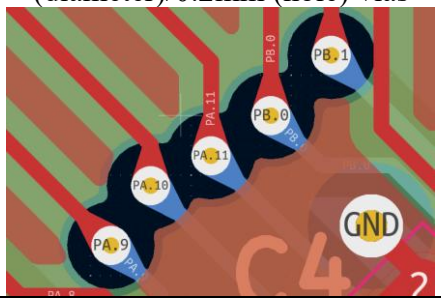


Component placement for PWM components

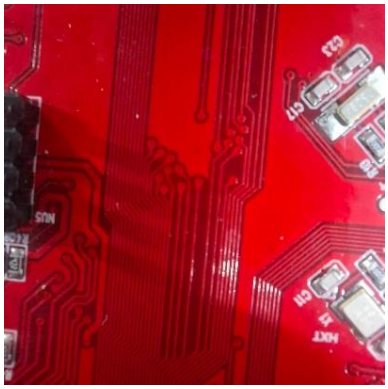
GPIO connectors are placed according to the operating voltage value of each interface. 5V and 3V3 interfaces are in their own area, each to make copper pour for each voltage rail simpler and continuous. All external connectors were moved to the PCB edges, which reduces routing congestion in the interior of the board and improves mechanical accessibility and cable management. The DAC circuitry was treated as a special case: although it is a low-voltage output from the MCU, it is routed through an op-amp stage that requires a 12 V reference to achieve the required output range, so it was placed in a mixed-signal area where both analogue 3.3 V and 12 V could be managed with careful local decoupling and reference routing. Because PWM uses both 5V and 3V3 for its logic shifter circuit, the area for 5V and 3V3 is alternated with sufficient distance between copper pours of each voltage rail

All relays were moved to the lower area of the board, powered by a dedicated 12 V rail, while their control signals operate on the 3.3 V low-voltage side. The 3.3 V domain is mainly routed on inner layers and restricted to sensitive logic and control circuitry. 12 V rail is routed along the bottom and board edges to keep high-current paths away from the MCU and communication interfaces. The MCU itself is placed near the centre and rotated diagonally, which shortens routing paths to the relay drivers without interfering with USB and Ethernet differential pairs. Finally, decoupling capacitors are placed strictly according to application notes, with VDDA, VREF, and related analogue supply pins grouped in a dedicated, well-decoupled area to minimise noise and improve ADC and DAC performance.

3.4. Teardrops on trace

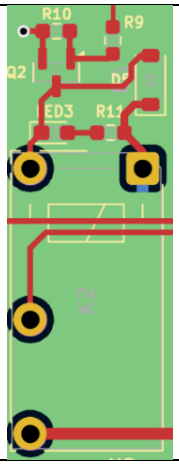
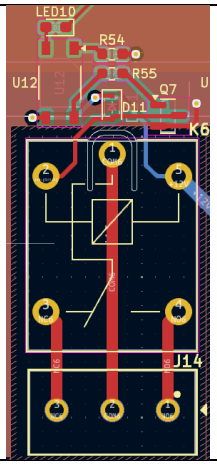
V1	V2
Not used because the minimum hole diameter requirement is 0.3mm	Used in traces connected 0.5mm (diameter)/0.2mm (hole) vias 

Teardrop is a feature where the junction of a trace, pad, or via width is gradually widened to form a teardrop-shaped connection. Teardrops serve as an additional copper area, strengthening the connection of a thin trace and/or small hole, pad, or via by providing gradual transition from a wider trace width to a narrower one to prevent mechanical stress. Teardrops also improve manufacturing tolerance by compensating for possible drill and layer misalignment, as the tapered copper shape provides a larger and more reliable connection area, thereby preserving both mechanical strength and electrical integrity even when slight positioning errors occur. In this PCB design, a teardrop is used for a trace of 0.2 mm width, which uses 0.5 mm (diameter)/0.2 mm (hole) via, taking reference from GPIO traces in Nuvoton NuMaker M467. The picture below is an example of traces with teardrops.



Example of traces with teardrops on Numaker M467 board

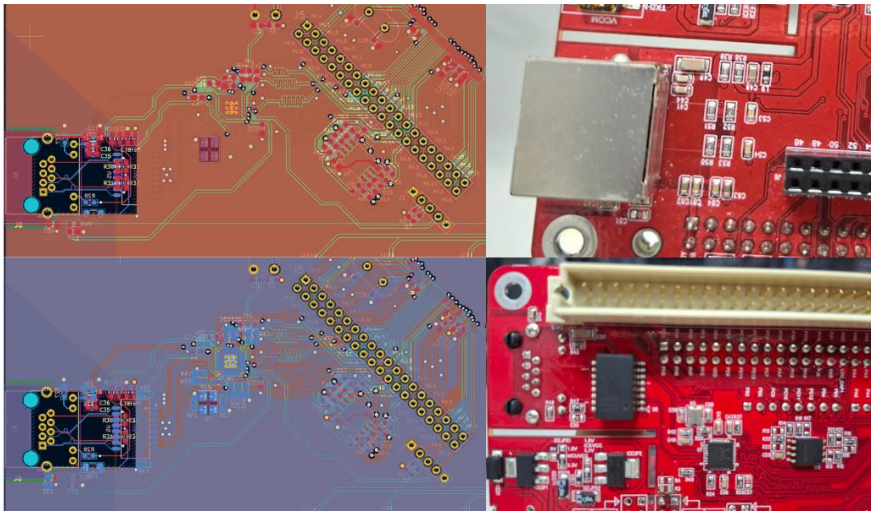
3.5. Relay

V1	V2
	

The relay layout in V2 introduces a significant configuration changes compared to V1 by changing the contact form into SPDT and integrating an optocoupler-based driver. In V1, the relay used is a SPST-NO type driven through BJT, resulting in limited isolation between the microcontroller and the high-voltage relay coil. The PCB V2 layout relocates the relay driver circuitry closer to the relay package and adds an optocoupler between the MCU GPIO and the coil drive transistor, providing galvanic isolation that prevents ground bounce, voltage spikes, and switching noise from coupling back into the sensitive digital domain. This isolation is particularly important in electrically noisy environments where inductive loads are switched, as it protects

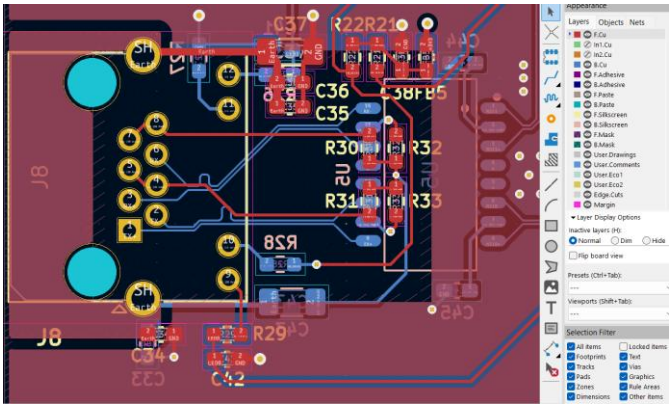
the microcontroller from transient overvoltage and electromagnetic interference. Furthermore, the SPDT relay in V2 enables greater wiring flexibility and functional versatility while the tighter routing and improved separation between control and power paths enhance overall reliability, safety, and maintainability of the relay subsystem.

3.6. Ethernet



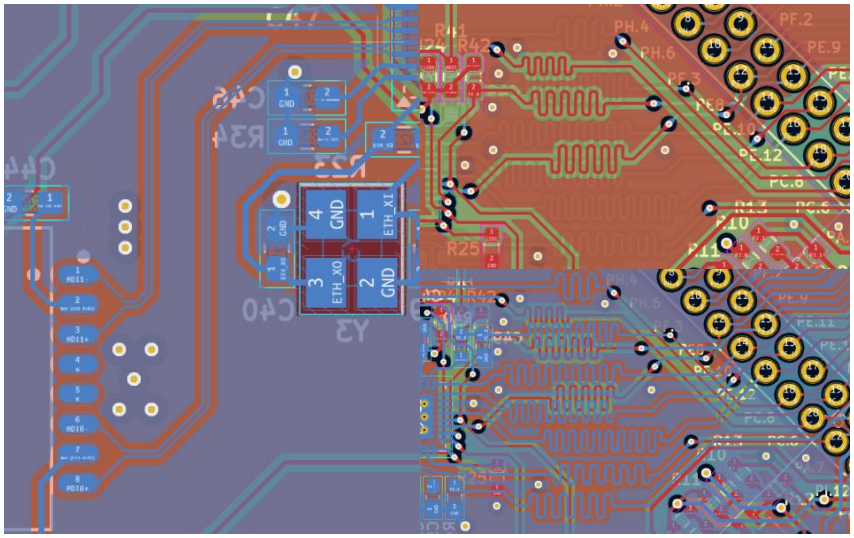
Left: Ethernet circuit in PCB layout; Right: Ethernet circuit in NuMaker M467

The component placement of Ethernet circuit refers to NuMaker M467 component placement. Referring to the layout shown, the MDI routing from the Ethernet port (RJ45) to the magnetics is treated as a cable-facing interface and therefore follows strict analogue and EMC-driven constraints.



MDI routing from RJ45 to magnetics (transformer)

Each MDI differential pair is routed as a tightly coupled 100 Ω differential pair, with minimal length, no stubs, and as few vias as possible to prevent discontinuities. Pair symmetry through the magnetics is preserved to maintain common-mode noise rejection. A clear keep-out region is kept beneath the RJ45 and transformer area, avoiding solid ground pours and high-speed digital traces to prevent parasitic coupling and to respect the isolation barrier between chassis/cable noise and the internal digital ground. Even though this segment uses differential routing, the priority here is EMI robustness and signal integrity at the physical cable interface rather than digital timing.



Left: MDI routing from magnetics (transformer) to PHY; Right: routing from PHY to MCU, with transition at extension pins, top picture is the top layer, and bottom is the bottom layer.

The routing from the magnetics to the PHY (MDI internal side) also uses differential pairs, but shifts focus to controlled reference planes and PHY timing requirements. These traces are routed over a continuous ground plane with controlled impedance of 50 ohm and careful via transitions, as this side is no longer directly exposed to cable-borne noise. Downstream of the PHY, the MII connection between the PHY and MCU transitions to single end type. The snla387 application notes by Texas Instruments state a maximum MII trace length of 6 inches to ensure timing margin; in this PCB, the routing adopts length-matching the MII signals to approximately 3 inches, a commonly referenced best-practice value for Ethernet MII designs. This tighter matching reduces skew between data and clock lines, improving setup/hold margins and overall link stability, while routing over a solid reference plane and keeping the signals isolated from noisy power and high-current regions.